



Reducing metabolizable protein supply: Effects on milk production, blood metabolites, and health in early-lactation dairy cows

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ABSTRACT

Our objective was to evaluate the effect of metabolizable protein (MP) supply on milk production, blood metabolites, and health in dairy cows during early lactation. Three experimental diets were formulated to contain 114, 107, 101 g of MP/kg of dry matter (DM; 114MP, 107MP, and 101MP, respectively) with crude protein contents of 17.0, 16.2, and 15.3% of DM, respectively. One hundred multiparous Holstein cows were fed 1 of these 3 diets during wk 1 to 3 and wk 4 to 13 of lactation in one of the following sequences: (1) 114MP and 107MP (114MP/107MP), (2) 114MP and 101MP (114MP/101MP), or (3) 101MP and 101MP (101MP/101MP). During wk 1 to 3, the 114MP and 101MP treatments were 20 and 27% deficient in estimated MP, respectively. From wk 4 to 13, the 114MP/107MP, 114MP/101MP, and 101MP/101MP treatments were 8, 12, and 13% deficient in estimated MP, respectively. Data were analyzed separately for wk 1 to 3, 4 to 13, and 1 to 13. Dry matter intake and energy-corrected milk (ECM) yield were not affected by treatment during wk 4 to 13 or wk 1 to 13; however, ECM yield decreased for 101MP versus 114MP from wk 1 to 3. Similarly, feed efficiency was not affected by treatment from wk 4 to 13 or wk 1 to 13, and was reduced with 101MP versus 114MP during wk 1 to 3. Milk N efficiency tended to increase for 101MP versus 114MP for wk 1 to 3 and increased with 101MP/101MP and 114MP/101MP relative to 114MP/107MP during wk 4 to 13 and wk 1 to 13. Treatment had no influence on yields and concentrations of milk components from wk 4 to 13 or wk 1 to 13; however, compared with 114MP, feeding 101MP tended to decrease milk fat yield and decreased yields of milk true protein and lactose for wk 1 to 3. Both milk and blood urea N concentra-

tions decreased for 101MP/101MP and 114MP/101MP relative to 114MP/107MP during wk 4 to 13 and wk 1 to 13, and were reduced with feeding 101MP versus 114MP from wk 1 to 3. Treatment had no effect on the incidence of diseases in cows throughout the study. Serum concentrations of total fatty acids, albumin, and aspartate aminotransferase did not differ between 101MP and 114MP; however, serum β -hydroxybutyrate concentration was lower in cows receiving 101MP during the first 3 wk of lactation. Compared with 114MP, feeding 101MP during wk 1 to 3 increased plasma concentrations of creatinine and 3-methylhistidine (3-MHis) but did not change the ratio of plasma 3-MHis to creatinine. We found no differences in plasma creatinine or the ratio of 3-MHis-to-creatinine among treatments from wk 4 to 13; however, 101MP/101MP and 114MP/101MP had elevated plasma 3-MHis compared with 114MP/107MP. Treatment had no effect on body weight and body condition score over the duration of the study. Collectively, despite reduced milk production for the first 3 wk of lactation, feeding the 101MP/101MP treatment sustained lactational performance and improved milk N efficiency without negatively affecting the frequency of diseases in dairy cows during the first 13 wk postpartum.

Key words: blood metabolite, metabolizable protein, production, transition cow

INTRODUCTION

Feeding high-protein diets is a common practice to maximize milk and milk protein yields in lactating dairy cows; however, overfeeding protein can result in decreased milk N efficiency, increased manure N excretion, and reduced profitability for dairy producers (Olmos Colmenero and Broderick, 2006; Yan et al., 2006; Chase et al., 2012). One promising approach to improve the efficiency of N utilization and reduce environmental N pollution is to feed low-protein diets to dairy cows (Chase et al., 2012; Hristov, 2016). For instance, fecal N and urinary N excretion have been found to decrease linearly as dietary CP content decreased from 18.4 to

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15.1% (Broderick, 2003) and from 19.4 to 15.0% (Olmos Colmenero and Broderick, 2006). However, inconsistent effects of lower-protein diets on milk and milk protein production in dairy cows have been observed.

A great deal of work has focused on early-lactation cows. Specifically, Law et al. (2009) reported that milk and milk protein yields decreased linearly from 35.4 to 25.4 kg/d and from 1.15 to 0.80 kg/d, respectively, when dietary CP decreased from 17.3 to 11.4%. Compared with a 19.3% CP diet, feeding a 15.4% CP diet reduced yields of milk and milk protein from 40.8 to 36.9 kg/d and from 1.18 to 1.09 kg/d, respectively; however, no differences in these 2 parameters were observed as dietary CP decreased from 19.3 to 17.4% (Wu and Satter, 2000). Moreover, yields of milk (mean = 36 kg/d) and milk protein (mean = 1.13 kg/d) did not differ for a 16.0 versus 19.7% CP diet (Hu et al., 2007). Likewise, no changes were detected in milk and milk protein yields between diets containing 17.6 versus 16.4% CP in cows with an average of 48 kg/d of milk (Wattiaux and Karg, 2004). Production responses of early-lactation cows to reduced protein supply may be dependent on dietary ingredient and nutritional composition, milk production, and physiological status of individual cows. To further investigate lower protein rations, it is critical to use MP for ration formulation, to take into account the effects of ruminal degradability and intestinal digestibility of protein sources (NRC, 2001; Chase et al., 2012).

Additionally, extra attention is required when formulating lower-protein diets for cows in early lactation, especially during the first few weeks after calving. During this period, depressed DMI makes it difficult to meet elevated nutrient demands (Grummer et al., 2004; McCabe and Boerman, 2020). Feeding lower-protein diets to fresh cows may cause severe mobilization of body protein, limit milk production, and increase the incidence of metabolic disorders (Bell et al., 2000; Ji and Dann, 2013; Amanlou et al., 2017). Plasma concentration of 3-methylhistidine (**3-MHis**), a biomarker of muscle mobilization, was greater in cows fed an MP-deficient diet (i.e., 13% deficiency) versus MP-adequate diets during the first 4 wk of lactation, although empty body CP loss was not altered (Tebbe and Weiss, 2021a,b). Similarly, plasma 3-MHis was higher for a diet with 15% MP deficiency compared with diets that were 5% deficient in MP during the first 3 wk after calving (Carder and Weiss, 2017). Additionally, reduced MP supply decreased yields of milk, ECM, and milk protein and fat (Tebbe and Weiss, 2021a), and yields of ECM and milk fat (Carder and Weiss, 2017). Overall, milk N efficiency and environmental impacts should be balanced with the risk of milk production losses and health disorders when developing protein

feeding strategies for early-lactation cows. We are not aware of any published studies evaluating MP supply in cows from parturition through 100 DIM.

The objective of this study was to investigate the effect of MP supply on DMI, milk yield and composition, blood metabolites, and disease incidence in dairy cows during early lactation. We hypothesized that, compared with feeding higher-MP diets for the first 13 wk of lactation, feeding higher-MP diets for the first 3 wk postpartum followed by lower-MP diets would sustain milk production and labile protein reserves while improving milk N efficiency in early-lactation cows.

MATERIALS AND METHODS

Experimental procedures were approved by the William H. Miner Agricultural Research Institute Animal Care and Use Committee (protocol no. 2012AUR04). The study was conducted at the Charles J. Sniffen Dairy Research Center of the William H. Miner Agricultural Research Institute (Chazy, NY) from March 2012 to January 2013.

Animal Management and Experimental Design

At 55 ± 3 d (mean $\pm$ SD) before expected parturition, cows were dried off and fed a far-off diet (12.6% CP, 57.2% NDF, 6.2% starch, and 1.34 Mcal/kg of NE_L; Table 1 and Supplemental Table S1, https://figshare.com/articles/online_resource/Supplemental_Tables_S1-4/15168840). Then, at 21 ± 3 d (mean $\pm$ SD) before expected calving, cows were switched to a close-up diet (15.0% CP, 44.2% NDF, 15.2% starch, and 1.46 Mcal/kg of NE_L; Table 1 and Supplemental Table S1). Dry cows were group-housed and fed through headlock feed barriers. Cows were trained to access feed from a Calan Broadbent Feeding System (American Calan Inc.) during the previous lactation or the far-off period. Following calving, cows were moved to a pen with sand-bedded freestalls and $\leq 100\%$ stall stocking density, fed individually once per day at 0830 h using the Calan Broadbent Feeding System, and milked 3 times daily at 0430, 1230, and 2030 h in a double-12 parallel milking parlor (Xpressway Parallel Stall System, Bou Matic LLC). Cows had free access to water via troughs available in pens.

Sample size was determined based on previous studies with transition cows completed at the William H. Miner Agricultural Research Institute, and the minimum number of replicates required to detect the effect of MP supply on DMI and milk yield was 20 per treatment (80% power and 95% confidence). Cows that delivered a single calf and appeared healthy were enrolled in a 13-wk completely randomized design experiment at

Table 1. Ingredient composition and chemical composition of the far-off and close-up diets (mean $\pm$ SD)

Item	Diet	
	Far-off	Close-up
Ingredient, % of DM		
Conventional corn silage	15.40	—
Brown midrib corn silage	—	45.07
Grass silage	78.10	17.41
Wheat straw	—	5.79
Concentrate mixture ¹	6.50	31.73
Nutrient composition, % of DM		
Samples, no.	9	10
DM, % as fed	37.2 $\pm$ 3.3	45.1 $\pm$ 1.9
CP	12.6 $\pm$ 0.8	15.0 $\pm$ 1.0
aNDF ²	56.8 $\pm$ 1.6	43.3 $\pm$ 3.5
ADF	36.9 $\pm$ 1.0	28.7 $\pm$ 1.8
NFC	22.1 $\pm$ 1.5	31.0 $\pm$ 2.7
Starch	6.2 $\pm$ 1.6	15.2 $\pm$ 1.7
Crude fat	3.1 $\pm$ 0.6	3.1 $\pm$ 0.3
Ash	7.2 $\pm$ 0.6	9.0 $\pm$ 1.1
NE _L , ³ Mcal/kg of DM	1.34 $\pm$ 0.03	1.46 $\pm$ 0.04

¹The far-off concentrate mixture contained 41.8% soybean meal, 26.6% high-protein mix (i.e., a blend of porcine blood and feather meal), 10.9% AminoPlus (Ag Processing Inc.), 5.2% soyhulls, 3.7% NaCl, 2.8% urea, 2.3% CaSO₄, 0.9% CaCO₃, 0.8% Rumensin (Elanco Animal Health), and 4.9% mineral and vitamin mix. The close-up concentrate mixture contained 22.3% canola meal, 12.9% soybean meal, 9.3% CaCO₃, 9.2% citrus pulp, 8.2% high-protein mix (i.e., a blend of porcine blood and feather meal), 7.4% dried distillers grains with solubles, 7.0% molasses, 3.7% wheat middlings, 1.4% MgO, 1.0% MgSO₄, 0.6% NaCl, 0.2% Rumensin, and 4.9% mineral and vitamin mix.

²aNDF = amylase and sodium sulfite-treated NDF.

³Estimated by Cumberland Valley Analytical Services (Hagerstown, MD).

calving. A total of 100 multiparous Holstein cows (2.82 $\pm$ 0.97 lactations; mean $\pm$ SD) were assigned in a balanced manner to 1 of 3 dietary treatments ($n = 33$ or 34 per treatment) according to calving date, parity, previous lactation milk yield, milk fat yield, milk protein yield, BW, and BCS. Three treatments were achieved by feeding 1 of 3 diets with different MP supply (i.e., 114, 107, and 101 g/kg of DM; **114MP**, **107MP**, and **101MP**, respectively) during wk 1 to 3 and wk 4 to 13 of lactation in one of the following sequences: (1) 114MP and 107MP (**114MP/107MP**), (2) 114MP and 101MP (**114MP/101MP**), or (3) 101MP and 101MP (**101MP/101MP**). Treatments were blinded to the research staff by coding the treatments with letters. The diets were composed of (DM basis) 52% forage (i.e., conventional corn silage, brown midrib corn silage, and grass silage) and 48% concentrate (i.e., ground corn, beet pulp, and concentrate mixture). Three concentrate mixtures were pelleted and delivered by Cargill Animal Nutrition and included in the diets to manipulate MP supply. The diets (i.e., 114, 107, and 101 g MP/kg of DM) contained different contents of CP (i.e., 17.0, 16.2, and 15.3% of DM, respectively)

and similar concentrations of NE_L ($\sim$ 1.65 Mcal/kg) and starch ($\sim$ 24.4% of DM).

The diets were formulated for early-lactation Holstein cows using the CPM-Dairy nutrition model (version 3.0; Cornell University, Ithaca, NY; University of Pennsylvania, Philadelphia, PA; and William H. Miner Agricultural Research Institute, Chazy, NY). The diets were expected to provide 80% ME and 75 to 90% MP requirements of a lactating dairy cow at 14 DIM with 748 kg of BW and 3.00 BCS, eating 19.2 kg of DMI/d, and producing 38.6 kg of milk/d with 3.8% fat and 3.1% true protein. The diets were also estimated to provide 100% ME and 90 to 110% MP requirements of a lactating dairy cow at 76 DIM with 705 kg of BW and 3.00 BCS, eating 28.2 kg of DMI/d, and producing 50.0 kg of milk/d with 3.8% fat and 3.1% true protein. Before the beginning of the study, the targeted MP supplies were 118, 108, and 99 g/kg of DM with CP contents of 17.5, 16.8, and 15.0% of DM, respectively. The targeted MP supplies were established based on the recommendations of field nutritionists and were commonly fed to early-lactation cows.

Dietary ingredients (Table 2; Supplemental Table S2, https://figshare.com/articles/online_resource/Supplemental_Tables_S1-4/15168840) were mixed and offered as TMR once daily using a Data Ranger mixer (American Calan Inc.), and orts were collected before feeding. Feed offered and orts were recorded by individual cows to calculate daily DMI. Milk yield was recorded using the ProVantage Information Management System (Bou Matic LLC). Body weight was measured using an Allweigh computerized scale (Allweigh Scale System Inc.), and BCS was assessed by 3 skilled individuals using a 5-point scale with 0.25 increments (Ferguson et al., 1994) within 24 h after calving and weekly thereafter until the end of the study.

Sample Collection and Analyses

Samples of forages, including conventional and brown midrib corn silages and grass silage, were collected thrice per week. Samples of concentrates (i.e., ground corn, beet pulp, and 3 concentrate mixtures), TMR, and orts were collected once per week. Following collection, 1 portion of these feed samples was dried in a forced-air oven at 105°C for 18 to 24 h for DM determination. The proportions of feed ingredients in TMR were adjusted on an as-fed basis when DM values of ingredients were outside the range of mean $\pm$ 1.2 SD. Dry matter values of TMR and orts were used to calculate DMI. Another portion of feed samples (i.e., forages, concentrates, and TMR) was stored at -20°C and then composited every 4 wk. The monthly composited samples were divided

Table 2. Ingredient composition (percentage of DM) of the diets fed to early-lactation cows

Item	Diet ¹		
	114MP	107MP	101MP
Conventional corn silage	20.07	20.07	20.07
Brown midrib corn silage	20.07	20.07	20.07
Grass silage	11.8	11.8	11.8
Ground corn	9.91	9.44	9.44
Beet pulp	4.72	4.72	4.72
Concentrate mixture			
Soybean meal	10.39	7.08	2.36
Canola meal	—	7.08	9.44
Wheat middlings	5.67	7.08	6.49
AminoPlus ²	7.08	2.36	—
Soybean hulls	5.55	5.55	5.55
DDGS ³	—	—	4.72
Molasses	1.18	1.18	1.77
Calcium carbonate	1.48	1.48	1.48
Megalac ⁴	0.59	0.59	0.59
Sodium sesquinate	0.56	0.57	0.57
Sodium chloride	0.43	0.43	0.43
Magnesium oxide	0.15	0.15	0.15
Magnesium sulfate	0.14	0.14	0.14
Vitamin E ⁵	0.06	0.06	0.06
AminoShure-M ⁶	0.03	0.03	0.02
Organic trace minerals ⁷	0.03	0.03	0.03
Inorganic selenium ⁸	0.03	0.03	0.03
Organic selenium ⁸	0.02	0.02	0.02
Mineral/vitamin mix ⁹	0.02	0.02	0.02
Vitamins A, D, and E ¹⁰	0.02	0.02	0.02

¹114MP = diet with 114 g of MP/kg of DM, 17.0% CP; 107MP = diet with 107 g of MP/kg of DM, 16.2% CP; 101MP = diet with 101 g of MP/kg of DM, 15.3% CP.

²AminoPlus is all soybean-based bypass protein source (Ag Processing Inc.).

³DDGS = dried distillers grains with solubles.

⁴Calcium salts of fatty acids containing 84.5% fat (Church and Dwight Co. Inc.).

⁵Contained 44,647 IU/kg vitamin E.

⁶AminoShure-M is a rumen-protected Met product with a lipid coating (Balchem Corp.).

⁷Contained 51,633 mg/kg Zn, 17,959 mg/kg Cu, 28,571 mg/kg Mn, and 3,673 mg/kg Co.

⁸Contained 606 mg/kg Se.

⁹Contained 0.02% Ca, 18.82% S, 153,737 mg/kg Zn, 30,303 mg/kg Cu, 136,364 mg/kg Mn, 3,384 mg/kg Co, 3,030 mg/kg I.

¹⁰Contained 30,497 kIU/kg vitamin A, 5,868 kIU/kg vitamin D, and 93,997 IU/kg vitamin E.

into 3 subsamples, with the first 2 subsamples shipped to Cumberland Valley Analytical Services (Hagerstown, MD) and the third subsample analyzed in-house. The first set was analyzed for CP (method 990.03; AOAC International, 2000), soluble CP (Krishnamoorthy et al., 1982), neutral detergent insoluble protein (Van Soest et al., 1991; Leco FP-528 Nitrogen Combustion Analyzer, Leco Corp.), RDP (Krishnamoorthy et al., 1983), NDF (Van Soest et al., 1991), ADF (method 973.18; AOAC International, 2000), ADL (Goering and Van Soest, 1970), starch (Hall, 2009), sugars (DuBois et al., 1956), crude fat (method 2003.05; AOAC International, 2006),

ash (method 942.05; AOAC International, 2000), and minerals (method 985.01; AOAC International, 2000). The second set was used for determination of 7-h in vitro starch digestibility of corn silages (i.e., conventional and brown midrib), ground corn, 3 concentrate mixtures and TMR, and fermentation analysis of forages. The third set (i.e., forages and TMR, no concentrates) was dried and ground using a Wiley mill (Arthur H. Thomas) to pass through a 1-mm screen, and then ground samples were analyzed for 24-h in vitro NDF digestibility (Goering and Van Soest, 1970).

Furthermore, another portion of TMR samples was used to determine particle size distribution on a DM basis (55°C) using dry vertical sieving (Ro-Tap testing sieve shaker model B, W. S. Tyler Combustion Engineering Inc.) with 19.0-, 13.2-, 9.5-, 6.7-, 4.75-, 3.35-, 2.36-, 1.18-, 0.60-, and 0.30-mm sieves for 10 min. The physical effectiveness factor of TMR was determined as the proportion of ration DM collected on ≥1.18-mm sieves (Mertens, 1997). The physically effective NDF of TMR was calculated as dietary NDF content multiplied by its physical effectiveness factor (Mertens, 1997).

Milk samples were collected using automatic samplers during 3 consecutive milkings weekly from d 8 to 91 postpartum, preserved with Bronolab-W II liquid preservative (D & F Control Systems Inc.), and stored at 4°C until shipped to Dairy One Cooperative Inc. (Ithaca, NY). Milk samples were analyzed for fat, true protein, lactose, MUN, and SCC using mid-infrared and flow cytometry procedures (Foss 4000, Foss Technology). Somatic cell score was calculated using the following equation: $SCS = \{[\log_{10}(SCC/1,000) - 2]/\log_{10}(2)\} + 3$. Yields of milk components were calculated using weekly milk yield (i.e., the sum of average milk yield at 0430, 1230, and 2030 h) and composited concentrations of milk components. Moreover, ECM yield was calculated using the following equation: $ECM = [(0.327 \times \text{kg of milk}) + (12.95 \times \text{kg of milk fat}) + (7.65 \times \text{kg of milk true protein})]$, with the ECM formula modified to account for the use of true protein instead of total protein (Tyrrell and Reid, 1965; Mark Stephenson, University of Wisconsin-Madison, <https://dairymarkets.org/PubPod/Reference/Library/Energy%20Corrected%20Milk>; Accessed Mar. 1, 2021). Feed efficiencies were calculated by dividing milk yield and ECM yield by DMI, and milk N efficiency was calculated as milk N output (i.e., milk true protein yield/6.38) divided by dietary N intake.

Blood samples (20 to 40 mL) were collected into BD Vacutainer SST tubes and K₂EDTA tubes (BD Diagnostics) via the coccygeal vein or artery before fresh TMR delivery at 2 wk before expected parturition, within 24 h after parturition, and on d 1, 3, 5, 7, 10, 13, 16, 19, 26, 33, 40, 47, 54, 61, 68, 75, 82, and 89

postpartum. Following sampling, SST tubes were allowed to clot for 30 min at room temperature for serum collection, and K₂EDTA tubes were kept on ice until being centrifuged for plasma collection. Blood samples were centrifuged at $1,200 \times g$ for 20 min at 20°C and 4°C to obtain serum and plasma samples, respectively, which were then stored at -20°C until further analysis. Serum samples obtained 2 wk before expected parturition and on d 1, 3, 5, 7, 10, 13, 16, 19, 26, 33, 40, 47, 54, 61, 68, 75, 82, and 89 postpartum were analyzed for urea N, and serum samples collected 2 wk before expected parturition and on d 1, 3, 5, 7, 10, 13, 16, and 19 postpartum were analyzed for total fatty acids (FA), BHB, albumin, and aspartate aminotransferase (AST) at the Animal Health Diagnostic Center (Ithaca, NY). Serum samples within 24 h after parturition were analyzed for calcium using a VetStat Electrolyte Blood Gas Analyzer (Idexx Laboratories Inc.). Plasma samples collected 2 wk before expected parturition and on d 1, 7, 13, 19, 26, 40, 54, 68 and 82 postpartum were shipped to Metabolic Technologies Inc. (Ames, IA) for determination of creatinine and 3-MHs.

Health status of individual cows was recorded throughout the study. The occurrence of all health disorders (i.e., metabolic, nonmetabolic, and infectious) was noted. Physical examinations of cows were conducted daily from d 1 to 14 postpartum and thereafter whenever needed based on visual observation of the cow or a decrease in DMI or milk yield. The physical examination included evaluation of attitude, rectal temperature, urine ketone concentration (Ketostix, Bayer Corp.), ruminal contractions, abdominal pings, vaginal discharge, and presence of fetal membranes.

Diet Evaluation and Statistical Analyses

After the completion of the study, the diets were evaluated using the Cornell Net Carbohydrate and Protein System (CNCPS; Van Amburgh et al., 2015) version 6.55, as the formulation and evaluation platform in AMTS.Cattle.Professional 4.16.1 (Agricultural Modeling and Training Systems LLC). Actual animal variables (i.e., lactation number, BW, BCS, DIM, milk yield and composition, and DMI) and feed variables (i.e., the nutritional composition of conventional and brown midrib corn silages, ground corn, and beet pulp), and the feed library values for the 20 ingredients in the concentrate mixtures were input. The chemical composition of wheat middlings and dried distillers grains with solubles were modified slightly so that the nutrient composition of the diets estimated by the AMTS model was similar to analyzed values as reported in Table 3. The estimated requirements and supplies of ME and MP across the treatments are presented in Table 4.

The diets were also evaluated using the NRC (2001) with the animal and feed information previously listed, and the NRC (2001) evaluation of the diets is shown in Supplemental Table S3 (https://figshare.com/articles/online_resource/Supplemental_Tables_S1-4/15168840).

The chemical composition of dietary ingredients and TMR and the particle size distribution of TMR were summarized using the MEANS procedure of SAS (version 9.4; SAS Institute Inc.) and reported as mean $\pm$ SD. In the present study, d 1 postpartum was defined as the first day when cows were milked 3 times. From d 1 to 91 postpartum, 16 cows were removed from the study due to the failure to use the Calan Broadbent Feeding System properly ($n = 7$), locomotion disorders ($n = 4$), hypocalcemia ($n = 1$), and failure to respond to medical treatment of mastitis ($n = 4$). Thus, 84 cows in total completed the study, including 28, 29, and 27 cows for the 114MP/107MP, 114MP/101MP, and 101MP/101MP treatments, respectively, and were included in the statistical analyses. Data were analyzed separately for wk 1 to 3, wk 4 to 13, and wk 1 to 13 of lactation, and tested for normality using the UNIVARIATE procedure and homogeneity of variance using the GLM procedure of SAS. Dry matter intake, milk yield, feed efficiency, milk N efficiency, milk composition, BW, and BCS were reduced to weekly values and analyzed using the MIXED procedure of SAS with week as repeated measure, according to the following model:

$$Y_{ijk} = \mu + C_i + W_j + T_k + W_j \times T_k + e_{ijk},$$

where Y_{ijk} = dependent variable, μ = overall mean, C_i = random effect of cow ($i = 1-84$), W_j = fixed effect of sampling week ($j = 1-3; 4-13; \text{ and } 1-13$), T_k = fixed effect of treatment ($k = 1-2; 1-3$), $W_j \times T_k$ = interaction between sampling week and treatment, and e_{ijk} = residual error. Moreover, changes in BW and BCS were analyzed with the same model, excluding sampling week and sampling week by treatment interaction.

Blood metabolites postpartum were analyzed using the MIXED procedure of SAS with day as repeated measure, according to the following model:

$$Y_{ijk} = \mu + C_i + D_j + T_k + D_j \times T_k + \text{Cov}_{ijk} + e_{ijk},$$

where Y_{ijk} = dependent variable, μ = overall mean, C_i = random effect of cow ($i = 1-84$), D_j = fixed effect of sampling day ($j = 1-19; 1-82$), T_k = fixed effect of treatment ($k = 1-2; 1-3$), $D_j \times T_k$ = interaction between sampling day and treatment, Cov_{ijk} = covariate variable for the i th cow within the j th day of the k th treatment, and e_{ijk} = residual error. The concentrations

Table 3. Chemical composition and in vitro digestibility (mean $\pm$ SD) of the diets fed to early-lactation cows (% of DM, unless otherwise noted)

Item	Diet ¹		
	114MP	107MP	101MP
Samples, no.	11	14	14
DM, % as fed	49.6 $\pm$ 0.8	48.7 $\pm$ 1.0	48.8 $\pm$ 0.7
CP	17.0 $\pm$ 0.8	16.2 $\pm$ 0.4	15.3 $\pm$ 0.3
Soluble CP, % of CP	40.7 $\pm$ 2.7	41.1 $\pm$ 2.4	41.2 $\pm$ 2.8
NDIP ²	1.9 $\pm$ 0.3	2.0 $\pm$ 0.2	2.2 $\pm$ 0.3
RDP ³	11.1 $\pm$ 0.2	10.8 $\pm$ 0.3	10.3 $\pm$ 0.3
RUP ⁴	5.9 $\pm$ 0.2	5.4 $\pm$ 0.3	5.0 $\pm$ 0.3
NDF ⁵	33.3 $\pm$ 1.0	34.4 $\pm$ 0.9	35.6 $\pm$ 1.2
ADF	19.5 $\pm$ 0.8	20.3 $\pm$ 0.7	20.9 $\pm$ 0.9
ADL	2.6 $\pm$ 0.3	3.0 $\pm$ 0.1	3.4 $\pm$ 0.3
NFC	40.3 $\pm$ 0.8	39.9 $\pm$ 0.9	39.6 $\pm$ 1.1
Starch	24.6 $\pm$ 0.9	24.5 $\pm$ 0.8	24.2 $\pm$ 0.8
Sugar	4.8 $\pm$ 0.4	4.5 $\pm$ 0.4	4.5 $\pm$ 0.4
Crude fat	3.3 $\pm$ 0.4	3.5 $\pm$ 0.4	3.9 $\pm$ 0.4
Ash	8.0 $\pm$ 0.5	8.0 $\pm$ 0.3	7.8 $\pm$ 0.4
Ca	1.18 $\pm$ 0.09	1.16 $\pm$ 0.09	1.22 $\pm$ 0.14
P	0.39 $\pm$ 0.03	0.42 $\pm$ 0.02	0.44 $\pm$ 0.02
Mg	0.37 $\pm$ 0.03	0.39 $\pm$ 0.02	0.41 $\pm$ 0.03
K	1.42 $\pm$ 0.07	1.34 $\pm$ 0.06	1.28 $\pm$ 0.05
S	0.24 $\pm$ 0.02	0.26 $\pm$ 0.01	0.28 $\pm$ 0.01
Na	0.43 $\pm$ 0.03	0.43 $\pm$ 0.03	0.41 $\pm$ 0.04
Cl	0.54 $\pm$ 0.03	0.55 $\pm$ 0.03	0.54 $\pm$ 0.04
Fe, mg/kg	277 $\pm$ 23	278 $\pm$ 26	287 $\pm$ 31
Cu, mg/kg	20 $\pm$ 3	20 $\pm$ 4	19 $\pm$ 2
Mn, mg/kg	76 $\pm$ 10	79 $\pm$ 13	75 $\pm$ 8
Zn, mg/kg	80 $\pm$ 11	83 $\pm$ 17	82 $\pm$ 8
24-h aNDF digestibility, % of aNDF ⁶	62.1 $\pm$ 0.9	60.7 $\pm$ 2.3	59.8 $\pm$ 1.6
7-h starch digestibility, % of starch	73.8 $\pm$ 3.4	74.6 $\pm$ 1.9	74.8 $\pm$ 3.3

¹114MP = diet with 114 g of MP/kg of DM, 17.0% CP; 107MP = diet with 107 g of MP/kg of DM, 16.2% CP; 101MP = diet with 101 g of MP/kg of DM, 15.3% CP.

²NDIP = neutral detergent insoluble protein.

³Measured using the procedure outlined by Krishnamoorthy et al. (1983).

⁴RUP% = CP% – RDP%.

⁵NDF with residual ash using α -amylase without sodium sulfite.

⁶aNDF = amylase and sodium sulfite-treated NDF.

Table 4. Cornell Net Carbohydrate and Protein System (CNCPS) evaluation¹ of the diets fed to early-lactation cows

Item	MP supply ²				
	Wk 1 to 3		Wk 4 to 13		
	114MP	101MP	114MP/107MP	114MP/101MP	101MP/101MP
NE _L , Mcal/kg of DM	1.65	1.63	1.60	1.61	1.61
MP, g/kg of DM	114	101	107	101	101
RDP, % of DM	10.3	9.2	9.3	8.8	8.8
RUP, % of DM	6.7	5.8	6.8	6.3	6.3
ME, Mcal/d					
Requirement	75.5	71.7	76.7	74.0	74.8
Supply	56.6	56.6	69.2	68.2	68.2
Balance	–18.9	–15.1	–7.5	–5.8	–6.6
MP, g/d					
Requirement	3,140	3,050	3,255	3,144	3,171
Supply	2,513	2,227	2,992	2,773	2,773
Balance	–626	–823	–263	–371	–398

¹All values were estimated using AMTS.Cattle.Professional 4.16.1 (Agricultural Modeling and Training Systems LLC), based on CNCPS version 6.55 with actual DMI, milk yield and composition, and BW of the cows, and feed nutritional composition throughout the study.

²114MP = diet with 114 g of MP/kg of DM, 17.0% CP; 107MP = diet with 107 g of MP/kg of DM, 16.2% CP; 101MP = diet with 101 g of MP/kg of DM, 15.3% CP. One of these 3 diets was fed during wk 1 to 3 and wk 4 to 13 of lactation in one of the following sequences: (1) 114MP and 107MP (114MP/107MP), (2) 114MP and 101MP (114MP/101MP), or (3) 101MP and 101MP (101MP/101MP).

of blood metabolites at 2 wk before expected parturition were used as the covariate terms and removed from the statistical model when $P > 0.05$.

Three covariance structures (i.e., compound symmetry, first-order autoregressive, and unstructured covariance) were tested, and the one with the lowest Akaike's information criterion was retained in the final model. Data from wk 1 to 3 were analyzed as an unbalanced repeated measures design (57 vs. 27 cows for 114MP vs. 101MP, respectively), and the MIXED procedure has the capacity to handle unbalanced data (Peterson et al., 2002; SAS Institute Inc., 2015). When an interaction of treatment by time was observed during wk 1 to 3 and wk 4 to 13, the SLICE option of SAS was used to compare treatments at each time point. Least squares means were separated using Tukey's procedure when a significant F -test ($P \leq 0.05$) was detected. All data above were reported as LSM $\pm$ SEM. Health data were analyzed by Fisher's exact test using the FREQ procedure of SAS with the exact option (Hatcher and Stepanski, 1994). Significance was declared at $P \leq 0.05$ and trends at $0.05 < P \leq 0.10$.

RESULTS AND DISCUSSION

Nutritional Composition of the Experimental Diets

Metabolizable protein supplies of 114, 107, and 101 g/kg of DM were achieved by decreasing the proportions of soybean meal and AminoPlus and increasing those of canola meal, wheat middlings, dried distillers grains with solubles, and molasses in the concentrate mixture (Table 2). Dietary contents of CP (17.0, 16.2, and 15.3%) and NDF (33.3, 34.4, and 35.6%) were different, and those of starch (24.6, 24.5, and 24.2%) and sugar (4.8, 4.5, and 4.5%) were similar across the diets (Table 3). The particle size distributions of the diets are shown in Supplemental Table S4 (https://figshare.com/articles/online_resource/Supplemental_Tables_S1-4/15168840). As expected, no differences were observed in the particle distributions of the diets. In addition, the physically effective NDF contents of the diets were similar (mean = 19.5%) and were slightly below the recommendation of 22 and 20% physically effective NDF for maintaining an average ruminal pH of 6.0 and a 3.4% milk fat content in early- to mid-lactation Holstein cows, respectively (Mertens, 1997).

According to the AMTS model, the estimated MP balance of cows fed the 101MP and 114MP diets averaged -823 and -626 g/d, respectively, for wk 1 to 3, and that of 101MP/101MP, 114MP/101MP, and 114MP/107MP averaged -398 , -371 , -263 g/d, respectively, for wk 4 to 13 (Table 4). Based on the overall efficiency of conversion of MP into net protein of 0.67

and 16.8% of empty BW as body protein (NRC, 2001; Tylutki et al., 2008), the average estimates of protein mobilization and BW losses in our cows from wk 1 to 13 were 26 and 178 kg, respectively. However, we observed that cows, on average, lost 49 kg of BW during wk 1 to 13 (discussed subsequently). The decrease in BW (-178 kg) calculated using the estimated MP balance was almost 4 times the measured one (-49 kg). This discrepancy may have been attributable to overestimation of MP requirements, underestimation of MP supply, or both, of early-lactation cows when using the CNCPS. The diets were also evaluated using the NRC (2001) software. The estimated MP balances of 101MP and 114MP during wk 1 to 3 were -616 and -443 g/d, respectively (Supplemental Table S3). In comparison, the MP balance estimated by the NRC (2001) software was -340 g/d for a 16.5% CP diet (i.e., 106 g of MP/kg of DM; Carder and Weiss, 2017) and -440 g/d for a 16.0% CP diet (i.e., 101 g of MP/kg of DM; Amanlou et al., 2017), in cows producing about 34 kg/d of milk during the first 3 wk of lactation. The more severe MP deficiency estimated using the NRC (2001) was mainly caused by higher milk yield (mean = 45.6 kg/d) in our cows. The MP estimates during wk 4 to 13 were similar between the CNCPS and the NRC (2001); however, the CNCPS overestimated MP requirements during the first 3 wk of lactation compared with the NRC (2001) software.

DMI and Lactational Performance

Dry matter intake, feed efficiency, and milk yield and composition are shown in Tables 5, 6, and 7. Both DMI and milk yield increased from wk 1 to 3 of lactation (week effect, $P < 0.001$; Table 5). In contrast, feed efficiencies expressed as milk yield and ECM yield per unit of DMI decreased for wk 1 to 3 (week effect, $P < 0.001$). During wk 4 to 13, DMI increased gradually, whereas ECM yield and feed efficiencies decreased steadily (week effect, $P < 0.001$; Table 6). In our study, cows reached their peak milk yield and ECM yield in wk 7 and 3, respectively (week effect, $P < 0.001$; Table 7).

During wk 1 to 3 of lactation, DMI (mean = 22.2 kg/d) was not affected by MP supply. Likewise, no differences in DMI were detectable from wk 4 to 13 and wk 1 to 13 of lactation. Consistent with our results, DMI did not change in cows when dietary CP content decreased from 18.4 to 16.3% (i.e., from 120 to 106 g of MP/kg of DM; Carder and Weiss, 2017) and from 19.7 to 16.0% (Hu et al., 2007). Conversely, DMI showed a linear decrease in response to decreasing dietary CP concentration from 18.4 to 15.1% (Broderick, 2003). Lee et al. (2011) reported that DMI was lower in dairy

Table 5. Dry matter intake, milk yield and composition, BW, and BCS in cows fed diets with different MP supply during lactation wk 1 to 3 of lactation

Item	MP supply ¹		SEM	P-value		
	114MP	101MP		Treatment	Week	Treatment × Week
DMI, kg/d	22.1	22.3	0.5	0.72	<0.001	0.03
Milk yield, kg/d	46.8	44.3	1.1	0.08	<0.001	0.20
Milk yield/DMI	2.12	1.99	0.05	0.05	<0.001	0.009
ECM yield	56.1	52.5	1.4	0.03	0.30	0.04
ECM yield/DMI	2.46	2.24	0.07	0.007	<0.001	0.05
Milk composition						
Fat, %	4.02	3.95	0.10	0.55	<0.001	0.20
Fat, kg/d	2.08	1.93	0.06	0.06	0.66	0.19
True protein, %	3.07	3.04	0.04	0.60	<0.001	0.25
True protein, kg/d	1.59	1.49	0.03	0.01	0.06	0.48
Lactose, %	4.85	4.81	0.03	0.31	<0.001	0.31
Lactose, kg/d	2.53	2.37	0.06	0.03	<0.001	0.98
SCS	2.01	1.84	0.30	0.64	<0.001	0.69
MUN, mg/dL	12.2	7.6	0.3	<0.001	0.03	0.76
BUN, mg/dL	11.7	7.7	0.2	<0.001	<0.001	<0.001
Milk N efficiency, %	43.9	45.8	0.9	0.08	<0.001	0.16
BW, kg	713	707	12	0.69	<0.001	0.87
BW change, kg/21 d	-51	-58	7	0.40	—	—
BCS	3.12	3.11	0.04	0.91	<0.001	0.60
BCS change/21 d	-0.16	-0.13	0.04	0.43	—	—

¹114MP = diet with 114 g of MP/kg of DM, 17.0% CP; 101MP = diet with 101 g of MP/kg of DM, 15.3% CP.

cows fed diets with 14.8 versus 16.7% CP (i.e., 97 vs. 108 g of MP/kg of DM, respectively). Previous studies have demonstrated that reduced dietary protein content (<15% CP or 9 to 10% RDP) can limit DMI by impairing rumen fermentation and nutrient digestibility in high-producing dairy cows (Hristov et al., 2004a;

Lee et al., 2011, 2012). The lack of differences in DMI across treatments in the current study indicated that ruminal bacterial growth and digestibility of nutrients were likely not affected by decreased RDP supply.

During wk 1 to 3 of lactation, milk yield tended ($P = 0.08$) to decrease by 2.5 kg/d, and ECM yield

Table 6. Dry matter intake, milk yield and composition, BW, and BCS in cows fed diets with different MP supply during wk 4 to 13 of lactation

Item	MP supply ¹			SEM	P-value		
	114MP/107MP	114MP/101MP	101MP/101MP		Treatment	Week	Treatment × Week
DMI, kg/d	27.9	27.4	27.4	0.5	0.70	<0.001	0.40
Milk yield, kg/d	53.9	51.4	53.2	1.3	0.37	<0.001	0.55
Milk yield/DMI	1.94	1.89	1.95	0.04	0.41	<0.001	0.56
ECM yield, kg/d	53.4	50.9	51.9	1.0	0.21	<0.001	0.70
ECM yield/DMI	1.92	1.87	1.91	0.03	0.38	<0.001	0.42
Milk composition							
Fat, %	3.50	3.49	3.42	0.09	0.77	<0.001	0.12
Fat, kg/d	1.85	1.79	1.81	0.04	0.48	<0.001	0.22
True protein, %	2.85	2.82	2.77	0.04	0.38	<0.001	0.46
True protein, kg/d	1.52	1.44	1.47	0.03	0.14	0.06	0.18
Lactose, %	4.89	4.90	4.87	0.03	0.77	0.003	0.69
Lactose, kg/d	2.63	2.52	2.60	0.06	0.47	<0.001	0.11
SCS	1.45	1.53	0.97	0.27	0.26	0.39	0.98
MUN, mg/dL	11.7 ^a	8.4 ^b	8.4 ^b	0.2	<0.001	<0.001	0.74
BUN, mg/dL	12.6 ^a	8.6 ^b	8.7 ^b	0.2	<0.001	<0.001	1.00
Milk N efficiency, %	35.3 ^b	37.6 ^a	38.5 ^a	0.5	<0.001	<0.001	0.53
BW, kg	695	691	691	11	0.97	<0.001	0.82
BW change, kg/70 d	7	5	1	5	0.70	—	—
BCS	2.98	2.87	2.92	0.05	0.29	0.002	0.90
BCS change/70 d	0.05	0.04	-0.01	0.04	0.59	—	—

^{a,b}Within a row, LSM without a common superscript differ ($P \leq 0.05$).

¹114MP = diet with 114 g of MP/kg of DM, 17.0% CP; 107MP = diet with 107 g of MP/kg of DM, 16.2% CP; 101MP = diet with 101 g of MP/kg of DM, 15.3% CP. One of these 3 diets was fed during wk 1 to 3 and wk 4 to 13 of lactation in one of the following sequences: (1) 114MP and 107MP (114MP/107MP), (2) 114MP and 101MP (114MP/101MP), or (3) 101MP and 101MP (101MP/101MP).

Table 7. Dry matter intake, milk yield and composition, BW, and BCS in cows fed diets with different MP supply during wk 1 to 13 of lactation

Item	MP supply ¹				P-value		
	114MP/107MP	114MP/101MP	101MP/101MP	SEM	Treatment	Week	Treatment × Week
DMI, kg/d	26.5	26.2	26.2	0.4	0.84	<0.001	0.35
Milk yield, kg/d	52.4	50.2	51.2	1.2	0.39	<0.001	0.17
Milk yield/DMI	1.99	1.94	1.97	0.04	0.62	<0.001	0.008
ECM yield, kg/d	54.0	51.5	52.1	1.0	0.21	<0.001	0.07
ECM yield/DMI	2.02	1.96	1.97	0.04	0.46	<0.001	0.006
Milk composition							
Fat, %	3.58	3.58	3.51	0.09	0.78	<0.001	0.24
Fat, kg/d	1.90	1.83	1.83	0.04	0.43	<0.001	0.15
True protein, %	2.89	2.86	2.82	0.04	0.41	<0.001	0.93
True protein, kg/d	1.54	1.46	1.48	0.03	0.12	<0.001	0.06
Lactose, %	4.88	4.89	4.86	0.03	0.71	<0.001	0.75
Lactose, kg/d	2.62	2.52	2.56	0.06	0.48	<0.001	0.02
SCS	1.48	1.67	1.10	0.26	0.29	<0.001	0.78
MUN, mg/dL	11.8 ^a	9.0 ^{bc}	8.3 ^{by}	0.2	<0.001	<0.001	<0.001
BUN, mg/dL	12.2 ^a	9.9 ^b	8.2 ^c	0.2	<0.001	<0.001	<0.001
Milk N efficiency, %	36.9 ^{by}	38.6 ^{abx}	39.8 ^a	0.5	<0.001	<0.001	<0.001
BW, kg	702	699	700	11	0.98	<0.001	0.99
BW change, kg/91 d	−49	−50	−59	8	0.59	—	—
BCS	3.05	2.93	2.98	0.05	0.23	0.11	0.99
BCS change/91 d	−0.30	−0.27	−0.27	0.06	0.93	—	—

^{a-c}Within a row, LSM without a common superscript differ ($P \leq 0.05$).

^{x,y}Within a row, LSM without a common superscript differ ($0.05 < P \leq 0.10$).

¹114MP = diet with 114 g of MP/kg of DM, 17.0% CP; 107MP = diet with 107 g of MP/kg of DM, 16.2% CP; 101MP = diet with 101 g of MP/kg of DM, 15.3% CP. One of these 3 diets was fed during wk 1 to 3 and wk 4 to 13 of lactation in one of the following sequences: (1) 114MP and 107MP (114MP/107MP), (2) 114MP and 101MP (114MP/101MP), or (3) 101MP and 101MP (101MP/101MP).

decreased ($P = 0.03$) by 3.6 kg/d, when cows were fed the 101MP diet compared with the 114MP diet (Table 5). It is important to note that cows receiving the 101MP diet had lower ECM yield in wk 2 but produced similar amounts of ECM in wk 3, compared with those offered the 114MP diet (interaction effect, $P = 0.04$). In contrast, yields of milk and ECM did not differ among the 101MP/101MP, 114MP/101MP, and 114MP/107MP treatments during wk 4 to 13 and wk 1 to 13 postpartum (Tables 6 and 7). Based on a meta-analysis by Hristov et al. (2004b), milk yield can be estimated by dietary concentrations of starch, fat, and MP, and DMI and DIM in dairy cows. Reduction in milk yield during wk 1 to 3 may be caused by a decrease in MP supply (−13 g of MP/kg of DM) with feeding the 101MP versus 114MP diet, as other impacting factors were similar between these 2 groups of cows. Tebbe and Weiss (2021a) reported that during d 1 to 25 postpartum, yields of milk and ECM averaged 32.5 and 39.7 kg/d in cows, respectively, and were reduced for a 16.9% CP diet with 87% MP requirements (i.e., 113 g of MP/kg of DM) compared with 20.1% CP diets containing an average of 107% MP requirements (i.e., 143 g of MP/kg of DM). In the study of Amanlou et al. (2017), compared with a 21.4% CP diet (i.e., 95% MP requirements or 138 g of MP/kg of DM), feeding a 16.0% CP diet (i.e., 78% MP requirements or 101 g of MP/kg of DM) reduced both milk yield (39.2 vs. 32.7

kg/d) and 4% FCM yield (36.1 vs. 31.4 kg/d) from calving until d 21 postpartum; however, no differences in these 2 parameters were observed between a 18.7% CP diet (i.e., 89% MP requirements or 120 g of MP/kg of DM) and the 21.4% CP diet. Taken together, it is conceivable that milk production can be limited in fresh cows when feeding diets with >10% MP deficiency, estimated by the NRC (2001).

Moreover, several studies have evaluated milk yield responses of early-lactation cows to decreasing dietary CP concentration achieved by replacing soybean meal with corn grains (Wu and Satter, 2000; Ipharraguerre and Clark, 2005) and reported that yields of milk and 3.5% FCM decreased as dietary CP decreased from 17.4 to 15.4% (mean = 38.2 kg/d of milk; Wu and Satter, 2000) and from 16.8 to 14.9% (mean = 41.0 kg/d of milk; Ipharraguerre and Clark, 2005). Conversely, neither milk yield nor 3.5% FCM yield was modified in early-lactation cows fed diets with 19.3 versus 17.4% CP (Wu and Satter, 2000), and 18.8 versus 16.8% CP (Ipharraguerre and Clark, 2005). These discrepant responses may be explained, at least partially, by differences in MP deficiency in cows across experiments (Daniel et al., 2016).

Feed efficiencies, expressed as milk yield and ECM yield per unit of DMI, were reduced ($P \leq 0.05$) during wk 1 to 3 postpartum when feeding the 101MP diet relative to the 114MP diet (Table 5). Specifically, feed-

ing the 101MP diet did not change milk yield/DMI in wk 1 but reduced this efficiency parameter during wk 2 and 3 of lactation (interaction effect, $P = 0.009$). An interaction effect ($P = 0.05$) between treatment and week for ECM yield/DMI was also observed, which resulted from a slower decrease in this parameter from wk 1 to 3, with feeding the 101MP versus 114MP diet. In contrast, feed efficiencies (i.e., milk yield/DMI and ECM yield/DMI) were not influenced by treatment during wk 4 to 13 (mean = 1.93 and 1.90 kg/kg, respectively; Table 6) and wk 1 to 13 (mean = 1.97 and 1.98 kg/kg, respectively; Table 7). Lower feed efficiencies with feeding the 101MP diet for wk 1 to 3 resulted from decreased milk yield and similar DMI compared with the 114MP diet.

Yields and concentrations of milk components, BUN, milk N efficiency, BW, and BCS are shown in Tables 5, 6, and 7. Yields and concentrations of milk components and milk N efficiency were influenced by time during wk 1 to 3 (week effect, $P \leq 0.06$; Table 5) and from wk 4 to 13 (week effect, $P \leq 0.06$; Table 6), with the exceptions of milk fat yield for wk 1 to 3 and milk SCS for wk 4 to 13. Additionally, BW and BCS decreased over the first 3 wk after calving (week effect, $P < 0.001$; Table 5). During wk 4 to 13, BW and BCS in dairy cows decreased first and then increased (week effect, $P < 0.001$; Table 6). An interaction effect ($P < 0.001$) of treatment by time was observed for BUN, which was caused by no treatment effect on BUN concentration on d 1, but decreased BUN concentration in cows fed the 101MP versus 114MP diet on other days during wk 1 to 3 of lactation. We detected no other interactions of treatment by week for milk composition, BUN, milk N efficiency, BW, and BCS during wk 1 to 3 and 4 to 13 (Tables 5 and 6).

During wk 1 to 3 of lactation, concentrations of milk components (i.e., fat, true protein, and lactose) were not influenced by MP supply (Table 5). In contrast, milk fat yield tended ($P = 0.06$) to decrease, and yields of milk true protein and lactose decreased ($P \leq 0.03$) for the 101MP diet relative to the 114MP diet. No differences in concentrations and yields of milk fat, true protein, and lactose were observed among treatments from wk 4 to 13 and wk 1 to 13 of lactation (Tables 6 and 7). Reductions in milk component yields from wk 1 to 3 were attributed to decreased milk yield (-2.5 kg/d) in cows receiving the 101MP diet as milk composition did not differ between the 101MP and 114MP diets. Changes in milk component yields with different dietary protein supplies were often driven by milk yield in lactating dairy cows (Wu and Satter, 2000; Ipharraguerre and Clark, 2005; Hu et al., 2007). Similar to our results, many previous studies did not observe

any changes in milk concentrations of fat, protein, and lactose when dietary CP content decreased from 17.3 to 16.1% CP (Broderick et al., 2008), and from 15.7 to 14.3% CP (Cabrita et al., 2011). By contrast, reducing dietary CP from 17.1 to 15.8% decreased both milk fat content and yield, which were likely caused by decreased apparent digestibility of NDF and ADF (Broderick et al., 2009). Decreased milk true protein concentration and yield were reported in dairy cows receiving a 15.1 versus 16.7% CP diet, which was likely a result of reduced DMI (Broderick, 2003).

Metabolizable protein supply had no effect on milk SCS in cows over the course of the study (Tables 5, 6, and 7). From wk 1 to 3 after calving, both MUN and BUN concentrations decreased ($P < 0.001$) for the 101MP versus 114MP diet (Table 5). Likewise, these 2 parameters were ($P < 0.001$) lower for the 101MP/101MP and 114MP/101MP treatments compared with the 114MP/107MP treatment from wk 4 to 13 (Table 6). During wk 1 to 13, the 101MP/101MP and 114MP/101MP groups had decreased ($P < 0.001$) MUN than the 114MP/107MP group, and 101MP/101MP cows tended ($P = 0.07$) to reduce MUN compared with 114MP/101MP cows (Table 7). The concentration of BUN for wk 1 to 13 was lowest for 101MP/101MP, highest for 114MP/107MP, and intermediate for 114MP/101MP. Reynolds and Kristensen (2008) has demonstrated that a greater amount of hepatic urea N recycles back to the rumen for microbial protein synthesis in dairy cows fed lower CP diets, thus reducing the concentrations of urea N in blood and milk.

Milk N efficiency tended ($P = 0.08$) to increase for the 101MP versus 114MP diet during wk 1 to 3 (Table 5) and was higher ($P \leq 0.003$) for the 101MP/101MP and 114MP/101MP treatments than the 114MP/107MP treatment during wk 4 to 13 (Table 6). During wk 1 to 13 of lactation, the 101MP/101MP group had greater ($P < 0.001$) milk N efficiency than the 114MP/107MP group, and the 114MP/101MP group tended ($P = 0.06$) to improve milk N efficiency than the 114MP/107MP group (Table 7). Nevertheless, it is critical to note that milk true protein yield decreased by 6% (i.e., 100 g/d) during the first 3 wk of lactation with feeding the 101MP diet. Comparable to our results, the efficiency of N use for milk production increased linearly in response to decreasing CP concentration from 17.3 to 11.4% (Law et al., 2009) and from 18.8 to 14.9% (Ipharraguerre and Clark, 2005). When feeding lower-protein diets to dairy cows, there is a greater amount of hepatic urea N recycling back to the rumen for microbial protein synthesis, which compensates for the reduction in dietary protein supply and thus improves the efficiency of N utilization (Reynolds and Kristensen, 2008). Alternatively, in the

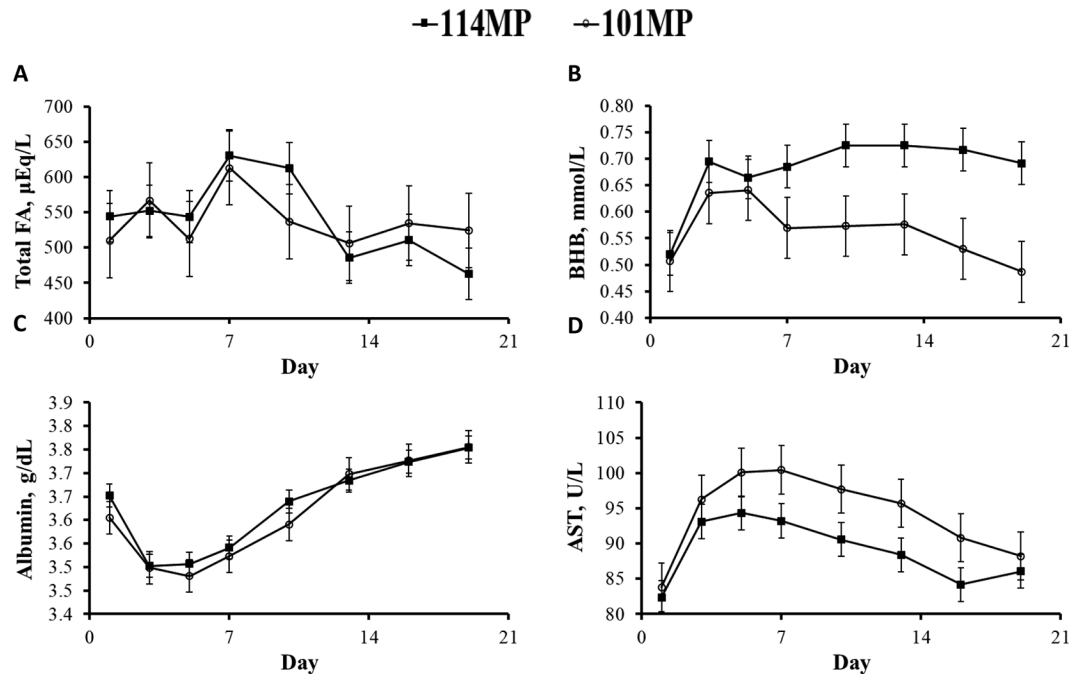


Figure 1. Serum concentrations of (A) total fatty acids (total FA; P -values: treatment = 0.91, day = 0.007, treatment $\times$ day = 0.67), (B) BHB (treatment = 0.02, day <0.001, treatment $\times$ day = 0.33), (C) albumin (treatment = 0.62, day <0.001, treatment $\times$ day = 0.63), and (D) aspartate aminotransferase (AST; treatment = 0.13, day <0.001, treatment $\times$ day = 0.64) in dairy cows from d 1 to 21 postpartum. 114MP = diet with 114 g of MP/kg of DM, 17.0% CP; 101MP = diet with 101 g of MP/kg of DM, 15.3% CP.

current study, improved N efficiency for lower MP diets (i.e., the 101MP vs. 114MP diet and the 101MP/101MP and 114MP/101MP vs. 114MP/107MP treatment) may be partly explained by enhanced mobilization of body protein as evidenced by increased plasma 3-MHis concentration. We observed that plasma 3-MHis was higher in cows receiving the 101MP versus the 114MP diet and the 101MP/101MP and 114MP/101MP versus the 114MP/107MP treatment (discussed subsequently). Circulating 3-MHis can be treated as a marker of muscle proteolysis in dairy cattle (Harris and Milne, 1981; Houweling et al., 2012). We did not observe any effects of MP supply on BW, BCS, and changes in BW and BCS throughout the study (Tables 5, 6, and 7). It is critical to note that, on average, our cows lost 54 kg of BW and 0.15 unit of BCS during the first 3 wk after calving, indicating the occurrence of negative energy balance in early lactation.

Metabolic Status, Protein Mobilization, and Health

We did not observe any interactions between treatment and sampling day on serum metabolites including total FA, BHB, albumin, and AST during d 1 to 21 postpartum (Figure 1). Serum total FA, BHB, albumin, and AST concentrations were modified from d 1 to 21

after calving (day effect, $P \leq 0.007$). Specifically, serum total FA increased acutely during wk 1 and decreased moderately during wk 2 and 3 (Figure 1A; $P = 0.007$). Serum BHB concentration increased first and then decreased gradually and reached peak values at d 5 and 10 for the 101MP and 114MP diets, respectively (Figure 1B; $P < 0.001$). Serum albumin decreased from d 1 to 5 and then increased thereafter (Figure 1C; $P < 0.001$), and serum AST increased sharply within the first 5 d postpartum and then gradually decreased (Figure 1D; $P < 0.001$). Moreover, MP supply had no effect on serum concentrations of total FA, albumin, and AST in dairy cows from d 1 to 21 postpartum (Figure 1A, 1C, and 1D); however, serum BHB concentration was lower ($P = 0.02$) for the 101MP versus the 114MP diet (Figure 1B).

Consistent with our results, plasma total FA concentration was not altered as dietary CP decreased from 17.5 to 14.2% (Christensen et al., 1994) and from 17.3 to 11.4% (Law et al., 2009). In our study, similar circulating total FA concentration between the 101MP and 114MP diets suggested that reduced protein supply did not enhance mobilization of adipose tissue for milk production, which was also confirmed by similar BCS changes during wk 1 to 3. Conversely, both serum total FA and BHB were elevated when dietary CP decreased

Table 8. Health issue events (n) in dairy cows fed diets with different MP supply during wk 1 to 13 of lactation

Item	MP supply ¹			P-value
	114MP/107MP	114MP/101MP	101MP/101MP	
Cows, no.	28	29	27	
Clinical disease				
Retained placenta	2	1	3	0.44
Displaced abomasum	0	1	0	0.99
Mastitis	6	3	3	0.50
Pneumonia	1	0	0	0.65
Foot and leg problems ²	11	10	12	0.76
Ketosis ³	1	3	1	0.61
Subclinical disease				
Ketonemia (≥ 1.4 mmol/L) ⁴	2	7	2	0.13
Ketonemia (≥ 1.0 mmol/L) ⁵	13	13	6	0.11
Subclinical hypocalcemia ⁶	1	0	1	0.54

¹114MP = diet with 114 g of MP/kg of DM, 17.0% CP; 107MP = diet with 107 g of MP/kg of DM, 16.2% CP; 101MP = diet with 101 g of MP/kg of DM, 15.3% CP. One of these 3 diets was fed during wk 1 to 3 and wk 4 to 13 of lactation in one of the following sequences: (1) 114MP and 107MP (114MP/107MP), (2) 114MP and 101MP (114MP/101MP), or (3) 101MP and 101MP (101MP/101MP).

²Included general lameness, abscess, pododermatitis circumscripta (ulcer), interdigital hyperplasia (corn), interdigital phlegmon (foot rot), digital dermatitis (heel wart), and white line disease.

³Ketosis was based on a urinary concentration of ketones ≥ 13.8 mmol/L or when cows showed signs of anorexia, ataxia, or abnormal behavior.

⁴Based on serum BHB concentration ≥ 1.4 mmol/L (Duffield et al., 2009).

⁵Based on serum BHB concentration ≥ 1.0 mmol/L (Ospina et al., 2010).

⁶Subclinical hypocalcemia was based on, within 24 h of parturition, serum calcium concentration ≤ 5.6 mg/dL.

from 19 to 16% (Amanlou et al., 2017), and plasma BHB increased in cows fed diets with 11.4 versus 17.3% CP (Law et al., 2009), which can be partly explained by reduced DMI in dairy cows fed lower-CP diets. Decreased DMI can further facilitate the release of total FA from adipose tissue and incomplete oxidation of FA via ketogenesis to meet energy demands in periparturient dairy cows (Drackley, 1999; Grummer, 2008). Despite similar DMI for the 101MP versus the 114MP diet, feeding the 101MP diet decreased plasma BHB concentration, which may have been associated with less energy requirements for lower milk production, and additional energy supply derived from enhanced protein mobilization. However, we did not observe any differences in the incidence of ketosis and subclinical ketonemia across treatments (Table 8). Blood albumin concentration has been proposed as an indicator of inflammatory status and liver function within 30 d postpartum (Bertoni et al., 2008). The cows used in our study had slightly higher plasma albumin (3.62 g/dL) than the group (3.50 g/dL) with the optimal liver function in the study of Bertoni et al. (2008), indicating that our cows had healthy liver condition within a few weeks after calving. According to Law et al. (2009), from d 1 to 150 postpartum, plasma albumin concentration was not changed between a 14.4 and 17.3% CP diet but decreased when CP decreased from 14.4 to 11.4%, which was likely a result of decreased protein supply for hepatic albumin synthesis (Raggio et al., 2007). Aspartate aminotransferase is an enzyme catalyzing the

reversible transfer of amino group between aspartate and glutamate, and blood AST concentration can also be used as a marker for liver function in periparturient dairy cows (Kida, 2002). No difference in serum AST concentration from d 1 to 21 postpartum was observed between the 101MP and 114MP diets, further confirming that feeding the 101MP diet to early-lactation cows did not impair liver function compared with the 114MP diet.

Plasma concentrations of creatinine and 3-MHis, and the ratio of plasma 3-MHis to creatinine throughout the study are shown in Figure 2. Plasma concentrations of creatinine and 3-MHis decreased sharply during the first 3 to 4 wk of lactation and then decreased slightly until the end of the study (day effect, $P \leq 0.01$; Figure 2A and 2B). The 3-MHis:creatinine ratio reached the peak value at approximately d 7, decreased sharply until around d 40, and remained stable for the duration of the study (day effect, $P < 0.001$; Figure 2C). An interaction effect ($P < 0.001$) between treatment and day was observed for plasma creatinine for d 1 to 21, which resulted from elevated plasma creatinine concentration on d 7, 13, and 19 but similar values on d 1 for the 101MP versus 114MP diet (Figure 2A). No other interaction effects of treatment $\times$ day were observed for these parameters.

Cows offered the 101MP diet had greater ($P \leq 0.03$) plasma creatinine and 3-MHis concentrations than those receiving the 114MP diet during the first 21 d postpartum (Figure 2A and 2B). The 101MP/101MP and

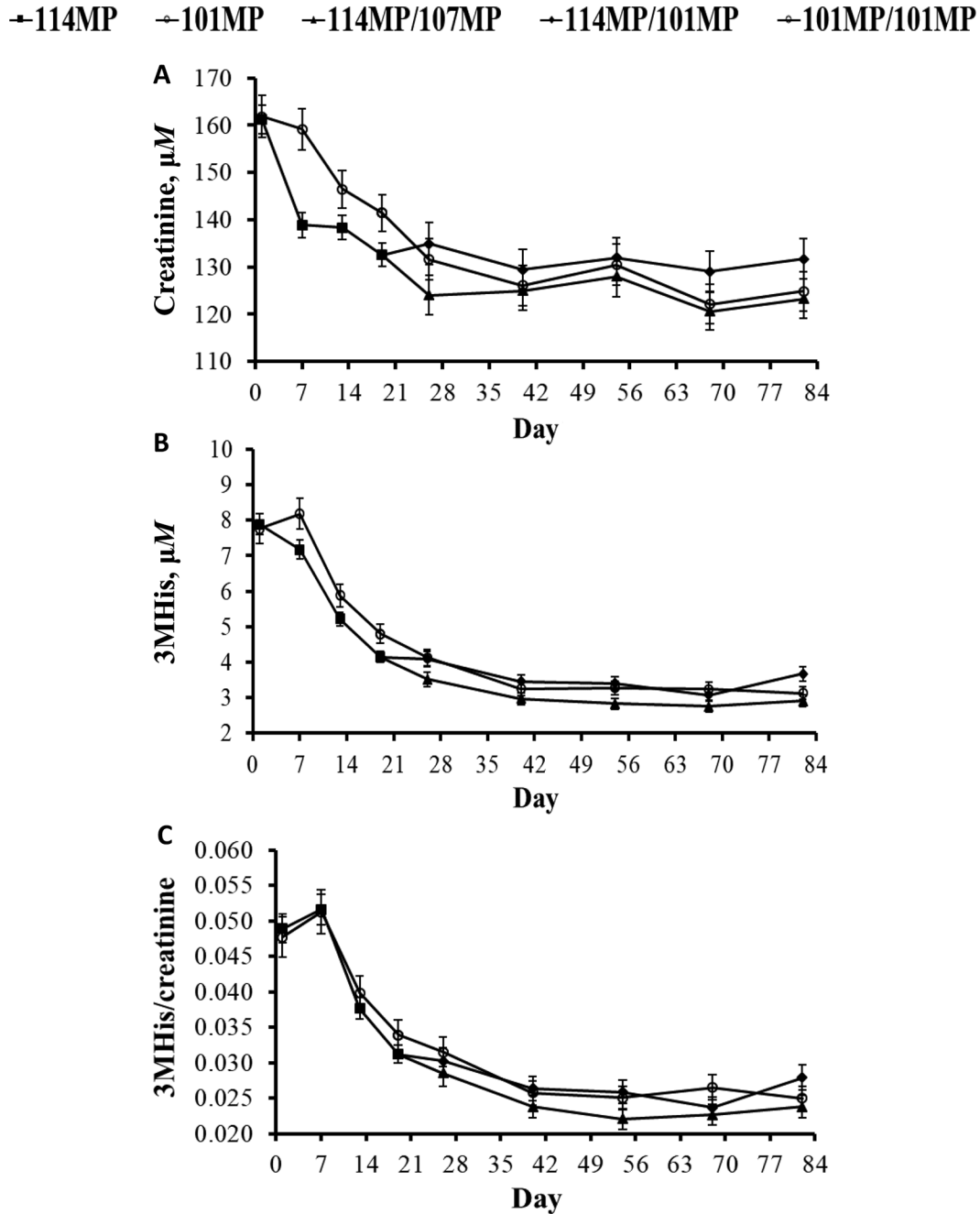


Figure 2. Plasma concentrations of (A) creatinine (P -values, d 1–21: treatment = 0.005, day <0.001, treatment $\times$ day <0.001; d 22–84: treatment = 0.31, day = 0.01, treatment $\times$ day = 0.87) and (B) 3-methylhistidine (3-MHis; d 1–21: treatment = 0.03, day <0.001, treatment $\times$ day = 0.14; d 22–84: treatment = 0.004, day <0.001, treatment $\times$ day = 0.50), and (C) ratio of plasma 3-MHis to creatinine (d 1–21: treatment = 0.58, day <0.001, treatment $\times$ day = 0.68; d 22–84: treatment = 0.21, day <0.001, treatment $\times$ day = 0.43) in dairy cows from d 1 to 84 postpartum. 114MP = diet with 114 g of MP/kg of DM, 17.0% CP; 107MP = diet with 107 g of MP/kg of DM, 16.2% CP; 101MP = diet with 101 g of MP/kg of DM, 15.3% CP. One of these 3 diets was fed during wk 1 to 3 and wk 4 to 13 of lactation in one of the following sequences: (1) 114MP and 107MP (114MP/107MP), (2) 114MP and 101MP (114MP/101MP), or (3) 101MP and 101MP (101MP/101MP).

114MP/101MP groups had higher ($P \leq 0.04$) plasma 3-MHis relative to the 114MP/107MP group from d 22 to 84 postpartum (Figure 2B). However, no differences

in the concentration of plasma creatinine among the 101MP/101MP, 114MP/101MP, and 114MP/107MP treatments were detected from d 22 to 84 postpartum

(Figure 2A). The ratio of plasma 3-MHis to creatinine was not changed by treatment either from d 1 to 21 or from d 22 to 84 (Figure 2C).

Creatinine is nonenzymatically converted from creatine and phosphocreatine at constant rates (1.1 and 2.6%/d, respectively), and more than 90% of body creatine is found in muscle tissues (Wyss and Kaddurah-Daouk, 2000). Thus, creatinine has been used as an indicator of total muscle mass in animals. An increase in circulating creatinine indicates muscle accretion, whereas a decrease in plasma creatinine indicates protein mobilization. A lower decrease in plasma creatinine between d 1 and 21 postpartum for the 101MP versus 114MP diet indicated that cows fed the 101MP diet may have mobilized less skeletal muscle during the first 3 wk of lactation, which did not align with MP balance and 3-MHis (discussed subsequently). This suggested that creatinine may not be a reliable biomarker of protein mobilization. According to Houweling et al. (2012), 3-MHis is produced in the catabolism of actin and myosin in skeletal muscles and has been considered a reliable indicator of muscle proteolysis in cattle (Harris and Milne, 1981; Chibisa et al., 2008). The increase in plasma 3-MHis with the 101MP diet from d 1 to 21 postpartum suggested that cows receiving the 101MP diet may have enhanced muscle degradation compared with those offered the 114MP diet. Likewise, elevated plasma 3-MHis concentration for the 101MP/101MP and 114MP/101MP groups during d 22 to 84 indicated that cows fed the 101MP/101MP and 114MP/101MP treatments may have mobilized greater amounts of body protein compared with cows receiving the 114MP/107MP treatment. Although changes in plasma creatinine concentration can be a method to monitor muscle mobilization in periparturient dairy cows, plasma creatinine concentration was correlated weakly with muscle depth measured using ultrasonographic measurement (Megahed et al., 2019). Compared with plasma creatinine concentration, monitoring changes in plasma 3-MHis concentration over time is a better way to estimate mobilization of body protein in dairy cows (McCabe and Boerman, 2020). Alternatively, elevated protein mobilization in cows receiving the 101MP diet was confirmed by lower negative MP balance for the 101MP diet (mean = -823 g/d) relative to the 114MP diet (mean = -626 g/d). Furthermore, the ratio of 3-MHis to creatinine can be used to compare proteolysis per unit of muscle mass in dairy cows (McCabe and Boerman, 2020). Similar 3MHis:creatinine ratios across treatments suggested that protein mobilization per unit of muscle mass was not influenced by MP supply in the present study. Treatment had no effect on the incidence of health issues, including retained placenta, displaced abomasum, mastitis, pneumonia, foot and leg prob-

lems, ketosis, ketonemia, and subclinical hypocalcemia throughout the study (Table 8).

CONCLUSIONS

Overall, reduced MP supply (i.e., from 114 to 101 g of MP/kg of DM) decreased milk production but did not alter DMI in cows during the first 3 wk of lactation. No differences in lactational performance were observed between the 101MP/101MP and 114MP/101MP treatments during wk 4 to 13 postpartum, indicating no carryover effects of feeding fresh cows lower-MP diets into later lactation. Despite elevated mobilization of body protein estimated by plasma 3-MHis for 101MP/101MP and 114MP/101MP cows versus 114MP/107MP cows, the incidence of diseases was not affected in early-lactation cows.

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